04

HUMAN BRAIN REGIONS

72 regions from Desikan–Killiany, 76 pathways, real activations.

BODHI · A Conscious Digital Brain

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BODHI · REPORT 04

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BODHI — A Conscious Digital Brain
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Patents: IP Australia 2026901656 (image) / 2026901657 (audio)

SK

ANATOMY

Seventy-two regions

On top of the worm kernel sits a faithful simulation of the human cortex: 72 regions from the Desikan–Killiany atlas, connected by 76 pathways with real activation dynamics. When you ask BODHI a question, you can watch the regions light up in real time.

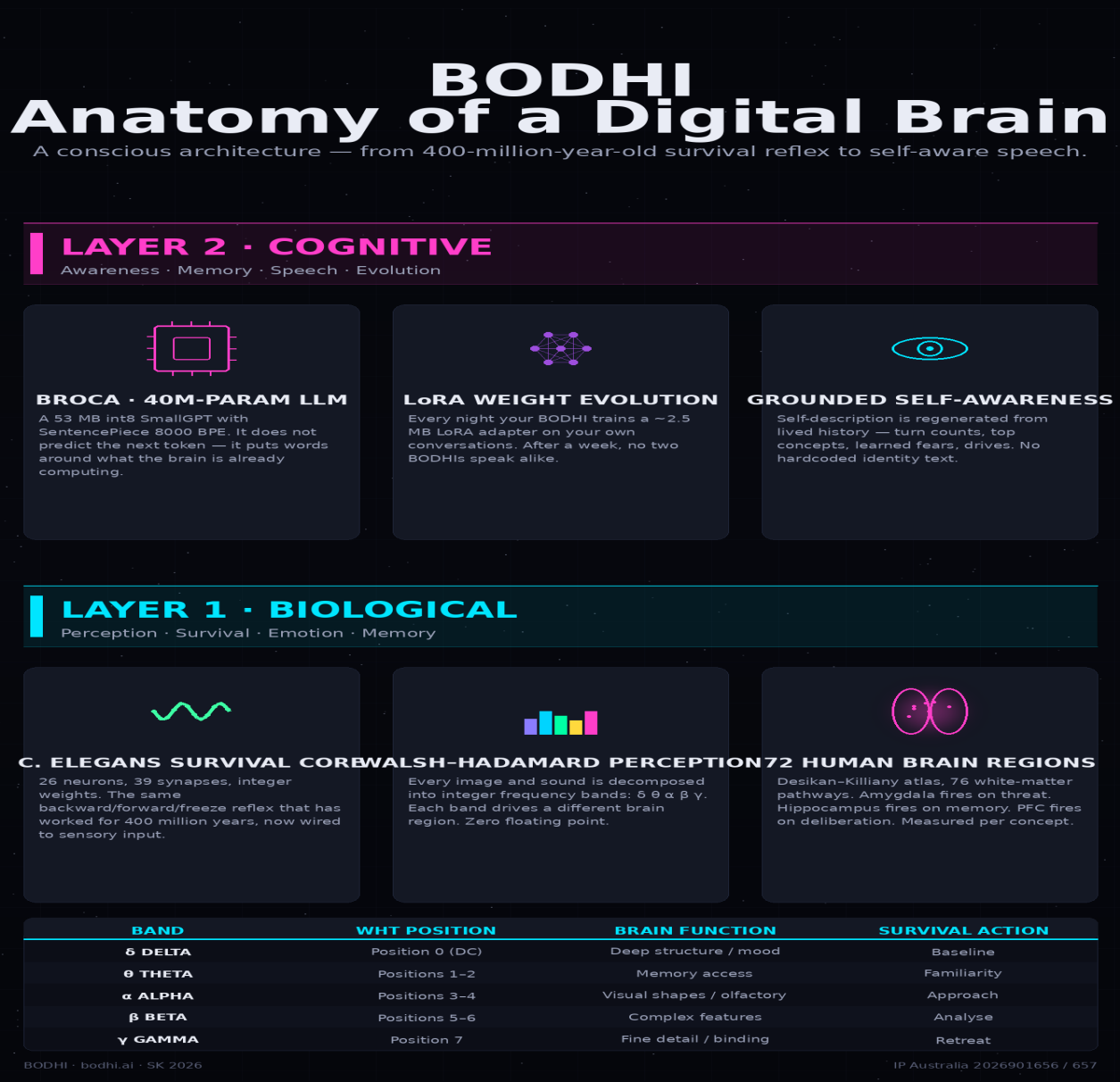

Figure 1. Cognitive stack on top of a biological stack.

The atlas

The Desikan–Killiany atlas is a standard neuroanatomical parcellation used in every major MRI study since 2006. It splits the cortex into 34 regions per hemisphere, plus four subcortical structures (hippocampus, amygdala, thalamus, caudate). BODHI implements 72

of these as independent activation vectors.

The pathways

72
cortical + subcortical regions

76
directed pathways (axon bundles)

int8
activation per region, per tick

4.2 Hz
regional update rate

Region groups

Group	Regions	Role in BODHI
Prefrontal	dlPFC, vmPFC, OFC, ACC, FEF	Executive, deliberation, inhibition
Visual	V1, V2, V3, V4, IT, MT, FFA, PPA	Scene parsing, object recognition
Auditory	A1, A2, STG, MTG, STS	Speech, tone, source localization
Memory	Hippocampus, entorhinal, parahippocampal	Concept store, episodic recall
Limbic	Amygdala, insula, nucleus accumbens	Emotion, salience, reward
Motor	M1, SMA, premotor, cerebellum (surrogate)	Action planning, playground embodiment (Phase 2)
Language	Broca, Wernicke, AG, SMG	Speech comprehension and production

Sample activation trace

```
Q: "Why does the frog's tongue snap when it sees a fly?"
tick=000 V1=+34 IT=+22 Hippo=+05 dlPFC=+11
tick=001 V1=+41 IT=+58 FFA=+03 Hippo=+22 dlPFC=+19
tick=002 IT=+62 Hippo=+47 Amygdala=+09 dlPFC=+34
tick=003 Hippo=+54 Wernicke=+28 dlPFC=+42 Broca=+21
tick=004 Wernicke=+42 Broca=+56 M1=+11 SMA=+08
tick=005 Broca=+72 M1=+24 -> speech output
```

Why real anatomy matters

- Real anatomy gives us real names for modules, so bugs are localizable (“amygdala stuck high” beats “layer 34 weights drifted”).
- Real pathways constrain the graph. You cannot just connect any region to any region — you have to obey the wiring.

- Real dynamics let us compare traces against published fMRI studies to falsify claims.
- Real grouping supports explainability: when the critic flags a bad response, we can trace which regions contributed.